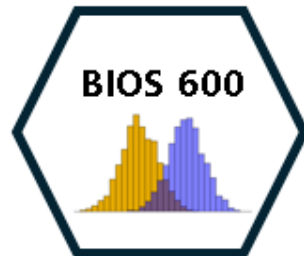


# Lecture 14: Non-Parametric tests

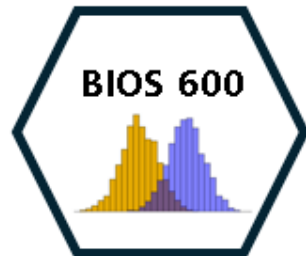
## BIOS 600 - Fall 2025

UNC - Chapel Hill



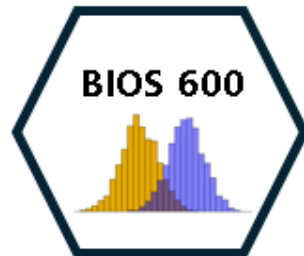
# Announcements

- ▶ Lab 6 due Friday 10/24 at 11:59pm.



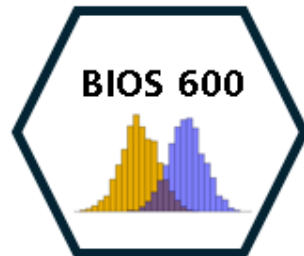
# Very optional reading

- ▶ P&G Chapters 13
- ▶ OI: Sections 8.1



# Overview

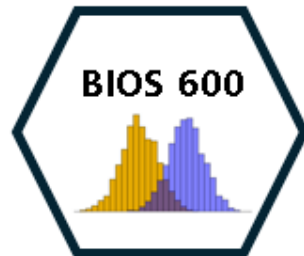
- ▶ Nonparametric methods
- ▶ Sign test, Wilcoxon signed-rank test, Wilcoxon rank-sum test
- ▶ Krustal-Wallis test



# Why nonparametric methods?

- ▶ For the methods we have studied so far, we have assumed the populations from which the data were drawn were either normally distributed or approximately so. This is a necessary property for the tests to be valid.
- ▶ Because the distributional form is assumed known, with only values of  $\mu$  and  $\sigma$  unknown, these methods are known as **parametric** methods. *t-test, ANOVA*
- ▶ **Nonparametric methods** make fewer assumptions about the nature of the underlying distributions and may be appropriate when some assumptions of parametric methods are not satisfied.

*small sample size*

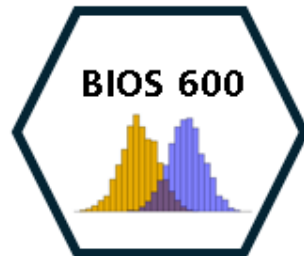


# Sign test

- ▶ The **sign test** is a nonparametric alternative to the paired t-test. It does not assume normality but just requires that observations are independent.
- ▶ The null hypothesis of the sign test is that the median difference among pairs in the underlying population is 0.

$(1, 2, 3, 4, 100)$        $\text{mean} \approx 27.5$

$1000$        $\text{median} = 3$



## Sign test: cold data

- ▶ A medical researcher claims that a new vaccine will decrease the number of colds in adults.
- ▶ You randomly select 14 adults and record the number of colds each has in a one-year period.
- ▶ After giving the vaccine to each adult, you again record the number of colds each has in a one-year period.
- ▶ At  $\alpha = 0.05$ , do the data support the researcher's claim?

*n = 14, does not meet rule of thumb*

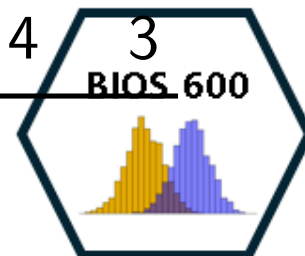
---

#  
Colds

---

|        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|--------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Before | 3 | 4 | 2 | 1 | 3 | 6 | 4 | 5 | 2 | 0 | 2 | 5 | 3 | 3 |
| After  | 2 | 1 | 0 | 3 | 1 | 3 | 3 | 2 | 2 | 2 | 2 | 3 | 4 | 3 |

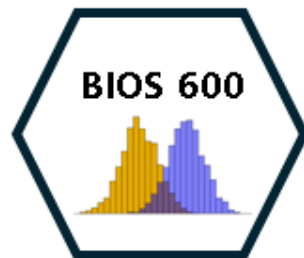
---



# Sign test

The sign test has just a few simple steps:

- ▶ Take the difference of each pair of observations (e.g., before-after)
- ▶ Record the sign of each difference (+, −, or 0)
- ▶ Count the number of + signs
- ▶ Test using the binomial distribution with  $n$  = number of pairs with nonzero differences and  $p = 0.5$ , as we would expect the **same number of positive and negative signs** under  $H_0$ .





# Sign test

The differences do NOT look very normal and we have a relatively small sample size. So, we cannot use previous methods to solve this question (and must rely on the sign test)!

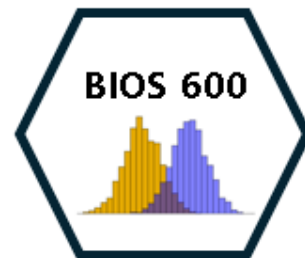
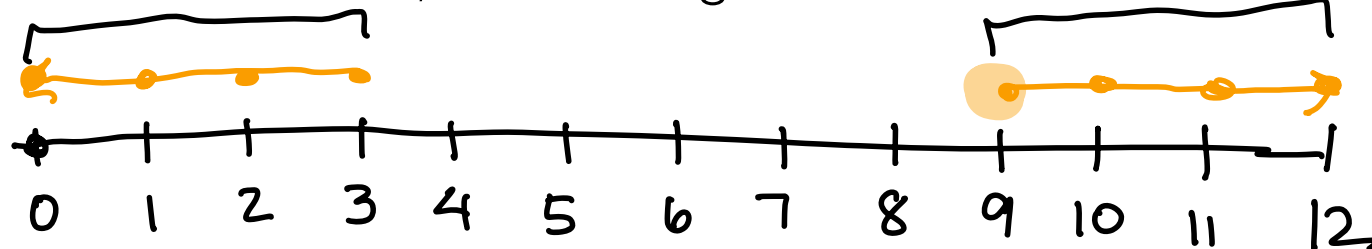
#  
Colds

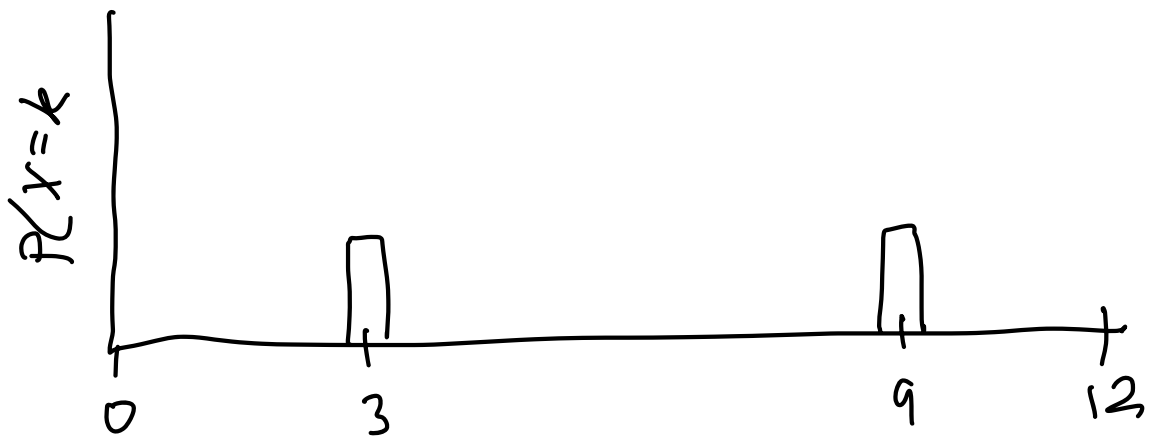
|        |   |   |   |    |   |   |   |   |   |    |    |   |   |   |
|--------|---|---|---|----|---|---|---|---|---|----|----|---|---|---|
| Before | 3 | 4 | 2 | 1  | 3 | 6 | 4 | 5 | 2 | 0  | 2  | 5 | 3 | 3 |
| After  | 2 | 1 | 0 | 3  | 1 | 3 | 3 | 2 | 2 | 2  | 3  | 4 | 3 | 2 |
| Diff.  | 1 | 3 | 2 | -2 | 2 | 3 | 1 | 3 | 0 | -2 | -1 | 1 | 0 | 1 |
| Sign   | + | + | + | -  | + | + | + | + | 0 | -  | -  | + | 0 | + |

×

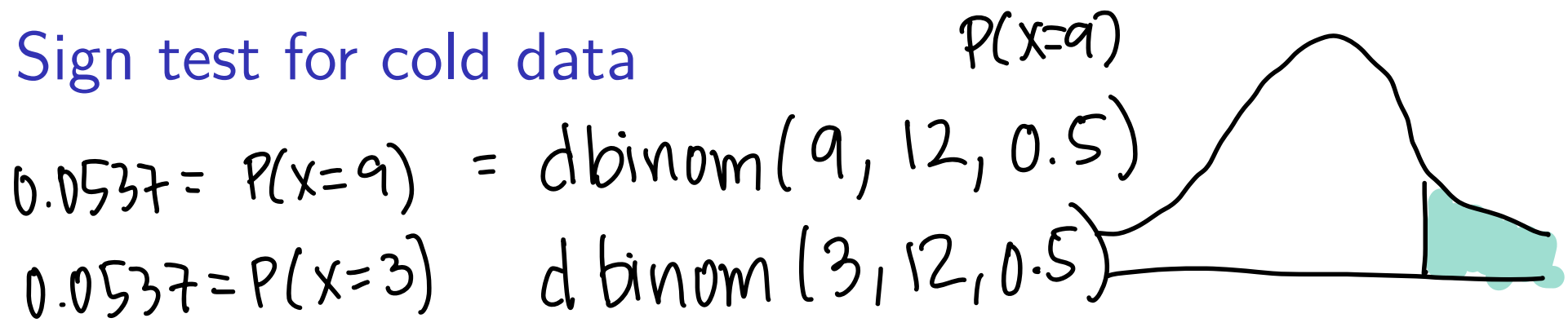
×

12 non-zero differences, with 9 + signs





## Sign test for cold data



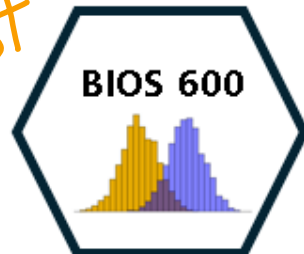
$H_0$ : Median difference in colds = 0 (i.e. same number of positive and negative signs)

$H_A$ : Median difference in colds  $\neq 0$  (i.e. different number of positive and negative signs)

- ▶ Remember, we had 9 successes out of 12 trials.
- ▶ What is the probability we see 9 successes out of 12 trials in a binomial distribution (or more extreme, *in either direction*), if the null hypothesis that  $p = 0.5$  is true?

proportion  
having increase or  
decrease in colds

two-sided  
test



# Conducting the sign test

= prob of "success" under  $H_0$

We can calculate, for  $X \sim \text{Binom}(12, 0.5)$ ,

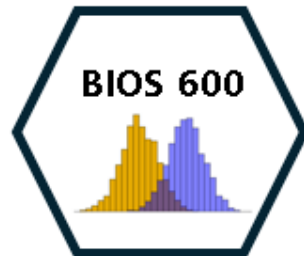
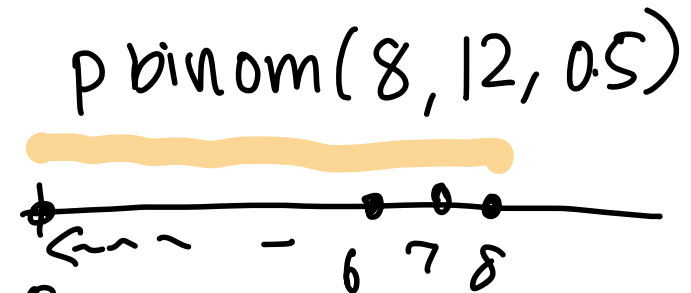
$$P(X = 9) + P(X = 10) + P(X = 11) + P(X = 12)$$

and then double it (for the two-sided test).

```
2*(1-pbinom(8, 12, 0.5))
```

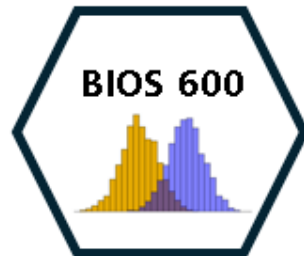
```
[1] 0.1459961
```

$$\text{pbinom}(3, 12, 0.5) + (1 - \text{pbinom}(8, 12, 0.5))$$



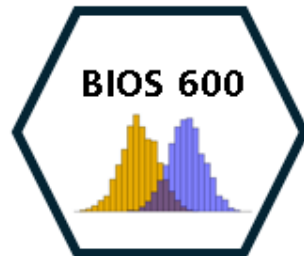
# Conducting the sign test

- **Conclusion:** With  $p = .145$ , we fail to reject  $H_0$  and hence conclude that there is not enough evidence to suggest there is a difference in the median number of colds in the years before and after vaccine administration.



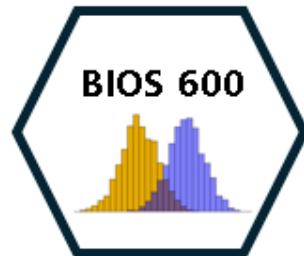
# Wilcoxon signed-rank test

- ▶ The sign test is appealing because it avoids all distributional assumptions. However, it ignores the magnitude of the differences.
- ▶ If we are willing to assume that the differences are symmetrically distributed around the median, we can incorporate the magnitude of the differences and gain considerable power using the **Wilcoxon Signed-Rank Test**.
- ▶ The Wilcoxon signed-rank test is an alternative to the paired t-test. The  $H_0$  is that the median difference in the underlying population is 0.



# Rank-based methods

- ▶ Many non-parametric methods have a similar flavor to parametric methods but are computed on the ranks instead of the observed data.
- ▶ The rank of an observation, among a set of observations, is its position when the observations are ordered from smallest to largest. The smallest observation has rank 1, next smallest has rank 2, and so forth.
- ▶ If observations are tied, the rank assigned to each is the average of the ranks appropriate to the equal numbers.



# Why ranks?

Using the ranks provides robustness against outliers:

► Weights: 175, 169, 190

► Ranks: 2, 1, 3

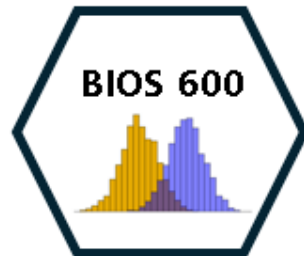
► Weights: 175, 169, 490

► Ranks: 2, 1, 3 (robust to outlier)

► Weights:  $\underset{1}{175}, \underset{2}{175}, \underset{3}{190}$

► Ranks: 1.5, 1.5, 3 (take sum of ranks and divide by number tied)

$$\frac{(1+2)}{2} = \frac{3}{2} = 1.5$$

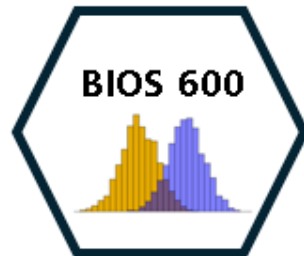




# Wilcoxon signed-rank test

Like the sign test, the **Wilcoxon signed-rank test** is for paired differences and has just a few simple steps:

- ▶ Calculate the difference for each pair of observations
- ▶ Rank the **absolute values** of these differences from smallest to largest (drop 0's; assign an average rank to ties); let  $n$  be the number of non-zero differences
- ▶ Assign each rank a  $+$  or  $-$  depending on the original sign of the difference
- ▶ Add up all positive ranks; add up all negative ranks; let  $T$  be equal to the smaller sum



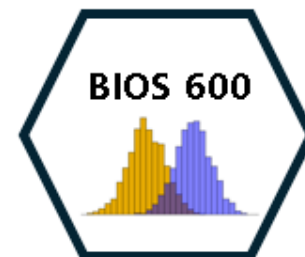
# Wilcoxon signed-rank test

Calculate the z-score

$$z_T = \frac{T - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}$$

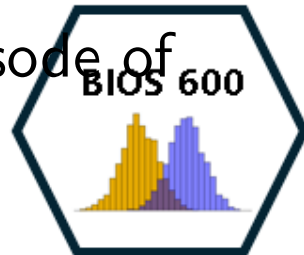
and obtain a p-value from the normal distribution.

Don't do this by hand – we will use R.



## Example: middle ear effusion

- ▶ A common symptom of otitis media in young children is the prolonged presence of fluid in the middle ear, known as **middle-ear effusion**. The presence of fluid may result in temporary hearing loss and interfere with normal learning skills in the first 2 years of life.
- ▶ One hypothesis is that babies who are breastfed for at least one month build some immunity and have less prolonged effusion than their bottle-fed counterparts.
- ▶ A small study of 14 babies is set up, with babies matched one-to-one by age, sex, socioeconomic status, and health condition. One member of the pair is a breastfed baby, and the other member is a bottle-fed baby. The outcome is the duration (days) of middle-ear effusion after the first episode of otitis media.



## Example: middle ear effusion

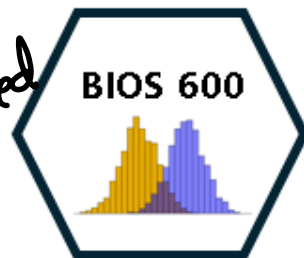
Duration of Effusion for Each Pair (Days):

| Pair   | 1  | 2 | 3  | 4  | 5 | 6  | 7  | 8  | 9 | 10 | 11 | 12 | 13 | 14 |
|--------|----|---|----|----|---|----|----|----|---|----|----|----|----|----|
| Breast | 26 | 3 | 12 | 28 | 7 | 39 | 12 | 30 | 7 | 15 | 65 | 10 | 19 | 11 |
| Bottle | 18 | 7 | 6  | 33 | 7 | 57 | 29 | 28 | 8 | 27 | 78 | 17 | 16 | 35 |

### **i** Question

What is  $H_0$ ?

$H_0$ : median difference in effusion duration between breast fed & bottle fed babies is 0



# Example: middle ear effusion

Duration of Effusion for Each Pair (Days):

|             | 1  | 2  | 3  | 4  | 5  | 6   | 7   | 8  | 9  | 10  | 11 | 12 | 13 | 14  |
|-------------|----|----|----|----|----|-----|-----|----|----|-----|----|----|----|-----|
| Breast      | 26 | 3  | 12 | 28 | 7  | 39  | 12  | 30 | 7  | 15  | 65 | 10 | 19 | 11  |
| Bottle      | 18 | 7  | 6  | 33 | 7  | 57  | 29  | 28 | 8  | 27  | 78 | 17 | 16 | 35  |
| Diff        | 8  | -4 | 6  | -5 | 0  | -18 | -17 | 2  | -1 | -12 | -  | -7 | 3  | -24 |
| Abs(Diff)   | 8  | 4  | 6  | 5  | 0  | 18  | 17  | 2  | 1  | 12  | 13 | 7  | 3  | 24  |
| Rank        | 8  | 4  | 6  | 5  | NA | 12  | 11  | 2  | 1  | 9   | 10 | 7  | 3  | 13  |
| Signed Rank | 8  | -4 | 6  | -5 | NA | -12 | -11 | 2  | -1 | -9  | -  | -7 | 3  | -13 |
|             |    |    |    |    |    |     |     |    |    |     | 10 |    |    |     |

13 non-zero ranks

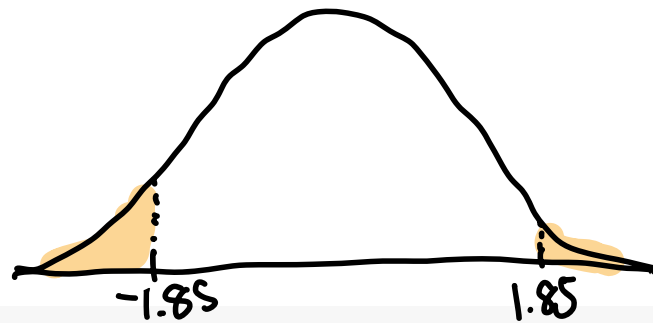
► Sum of positive ranks is smaller than sum of negative ranks

► Sum of positive ranks =  $8 + 6 + 2 + 3 = 19 = T$

►  $z_T = \frac{19 - \frac{13(14)}{4}}{\sqrt{\frac{13(14)(27)}{24}}} = \frac{19 - 45.5}{3} = -1.85 \rightarrow$  Now, let's calculate our p-value.



## Calculating the p-value



```
zt <- (19 - (13 * 14 / 4)) / sqrt(13 * 14 * 27 / 24)
```

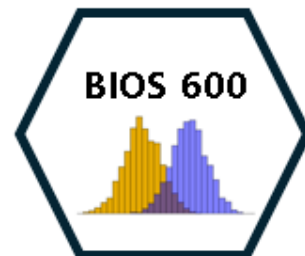
```
zt
```

```
[1] -1.85197
```

```
pnorm(zt) * 2 # two-sided test
```

```
[1] 0.06403015
```

- **Conclusion:** With  $p > 0.05$ , we fail to reject the null hypothesis. We do not have sufficient evidence to suggest that there is a difference in the median of the paired differences in effusion duration between being breast and bottle-fed.

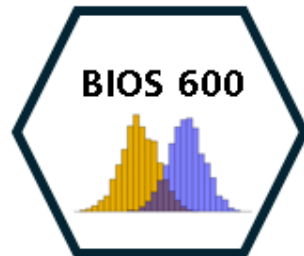


# Wilcoxon rank sum test

Not to be confused with the Wilcoxon Signed-Rank Test!

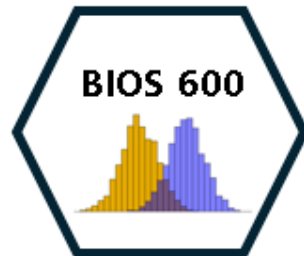
The **Wilcoxon Rank Sum Test**:

- ▶ Is the non-parametric counterpart of the two-sample t-test
- ▶ Assumes two samples **are independent** but does not require normality or equal variance of groups
- ▶ Assumes two distributions have “roughly the same shape”
- ▶ Also called the **Mann-Whitney U test** or **Wilcoxon-Mann-Whitney test**



# Mann-Whitney U test

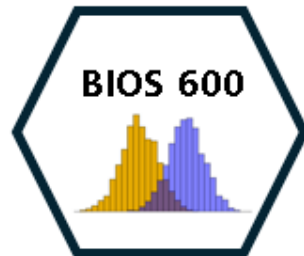
- ▶ The Mann-Whitney U-test is an extension of the Wilcoxon signed rank test for **independent samples**.
- ▶ Instead of taking differences, the observations are lumped together and ranked, with then the sum of the ranks calculated by group.
- ▶ We'll use software to perform this test.





# Example: Exercise capacity and coronary artery disease

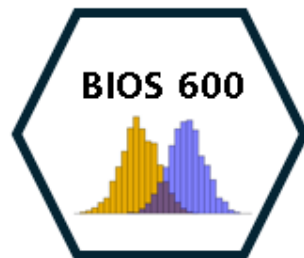
- ▶ Consider data from a two-group study of exercise capacity.
- ▶ Two groups of men, one with diagnosed three-vessel coronary artery disease (3VD), and the other group of men with suspected disease (SD) in one or more vessels.
- ▶ The total time participants could exercise on a treadmill set to increase in speed and slope according to a set schedule, is below:
- ▶ 3VD times: 864, 636, 638, 708, 786, 600, 1320, 750, 594, 750
- ▶ SD times: 1014, 684, 810, 990, 840, 978, 1002, 1110
- ▶ 3VD median: 729
- ▶ SD median: 984



## Example: Exercise capacity and coronary artery disease

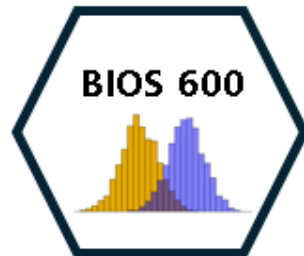
| Value | Rank | Group | Value | Rank | Group |
|-------|------|-------|-------|------|-------|
| 594   | 1    | 3VD   | 810   | 10   | SD    |
| 600   | 2    | 3VD   | 840   | 11   | SD    |
| 636   | 3    | 3VD   | 864   | 12   | 3VD   |
| 638   | 4    | 3VD   | 978   | 13   | SD    |
| 684   | 5    | SD    | 990   | 14   | SD    |
| 708   | 6    | 3VD   | 1002  | 15   | SD    |
| 750   | 7.5  | 3VD   | 1014  | 16   | SD    |
| 750   | 7.5  | 3VD   | 1110  | 17   | SD    |
| 786   | 9    | 3VD   | 1320  | 18   | 3VD   |

- ▶ Sum of ranks in SD group,  $n = 8$ : 101
- ▶ Sum of ranks in 3VD group,  $n = 10$ : 70



# Example

- ▶ The sum of the ranks in the SD group with  $n = 8$  was 101, and the 3VD group had 10 subjects with sum of ranks 70. Is this enough evidence to reject the null hypothesis?



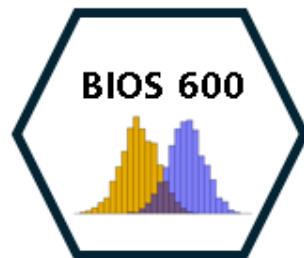
# Mann-Whitney U Test Statistic

| Group | n  | Sum of Ranks |
|-------|----|--------------|
| SD    | 8  | 101          |
| 3VD   | 10 | 70           |

**Step 1** Compute U-statistics.

$$U_1 = R_1 - \frac{n_1(n_1 + 1)}{2}, \quad U_2 = R_2 - \frac{n_2(n_2 + 1)}{2}$$

where  $R_1$  and  $R_2$  are the sums of ranks in groups 1 and 2, respectively, and  $n_1$  and  $n_2$  are group sample sizes.

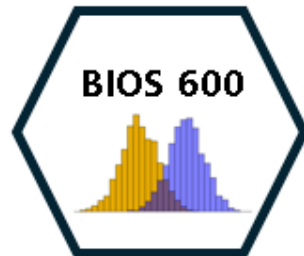


# U-statistic

- ▶ When both sample sizes are  $\geq 8$ , the sampling distribution of  $U$  is approximately normal with mean and standard deviation:

$$\text{mean}_U = \frac{n_1 n_2}{2}, \quad \text{sd}_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$$

**No need to memorize this!**

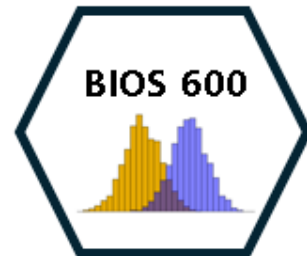


# In R

```
# Given data  
n1 <- 8; n2 <- 10  
R1 <- 101; R2 <- 70  
  
U1 <- R1 - n1 * (n1 + 1) / 2  
U2 <- R2 - n2 * (n2 + 1) / 2  
U1; U2
```

```
[1] 65
```

```
[1] 15
```



# In R

```
mean_U <- n1 * n2 / 2
sd_U <- sqrt(n1 * n2 * (n1 + n2 + 1) / 12)
z <- (U1 - mean_U) / sd_U
p_value <- 2 * (1 - pnorm(z))

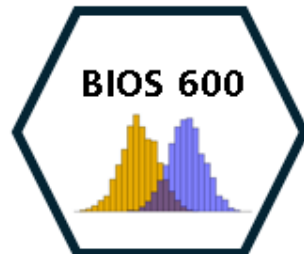
z; p_value
```

```
[1] 2.221308
```

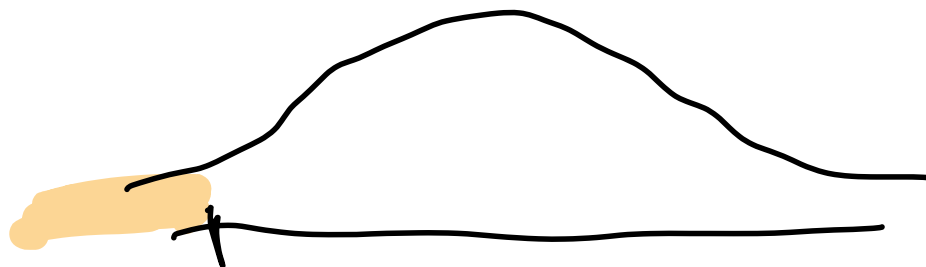
```
[1] 0.02633009
```



- As it turns out, we reject  $H_0$  and conclude the two groups do not have the same median. The median exercise time in the SD group is higher compared to the 3VD group.



# Note



💡 Alternative way of calculating p-value

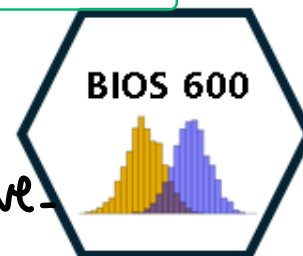
▶ Note that we can also calculate our p-value using:

```
2 * pnorm(-abs(z))
```

```
[1] 0.02633009
```

- ▶ Why can we do this? Can you draw a picture to explain why these two methods yield the same result? (Look up ?abs in the Console)
- ▶ Why might this code be **more generalizable** compared to the previous code to calculate the p-value?

Regardless of  $z$  pos. or negative,  
place  $z$  on negative side of  $N(0,1)$  curve.





# Kruskal-Wallis test

t test  $\rightarrow$  Anova

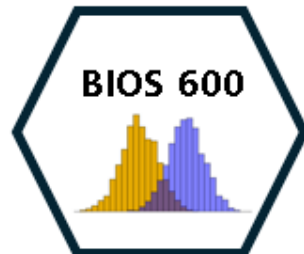
2 groups

3+ groups

- ▶ The **Kruskal-Wallis Test** is the nonparametric version of ANOVA, generalizing the Wilcoxon rank sum test to more than 2 groups.

## Kruskal-Wallis test for pet data

- ▶ Consider the study of pulse rate from the ANOVA lecture.
- ▶ Remember that pulse rates were not necessarily normally distributed within groups.



# Kruskal-Wallis test for pet data

- ▶ Calculating the sum of the ranks, we find that in the pet group, out of  $n = 30$ , the sum of the ranks was 863; out of the  $n = 30$  in the friend group, the sum of the ranks was 1797, and out of the  $n = 30$  where neither were present, the sum of the ranks was 1435. Is this enough evidence to reject the null hypothesis? (Ask software).

```
kruskal.test(pulse ~ group, data = pulse_data)
```

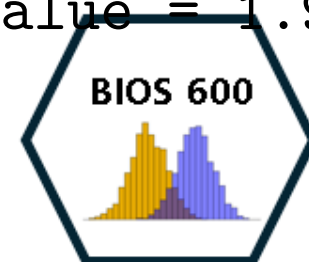
Kruskal-Wallis rank sum test

data: pulse by group

Kruskal-Wallis chi-squared = 21.662, df = 2, p-value = 1.97e-05

The p-value is < 0.0001. What may we conclude?

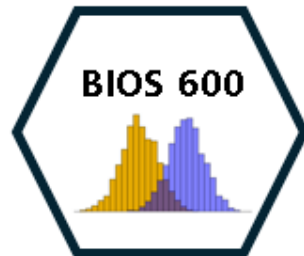
= # groups - 1



## Pairwise comparisons: pet vs. friend

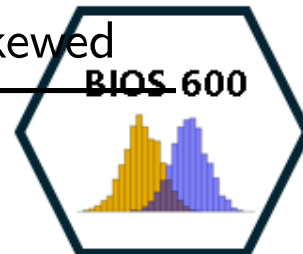
- ▶ Again, we can step-down and look at pairwise differences. In this case, we would use the Wilcoxon Rank-Sum test (why?).

If you start w/ nonparametric,  
keep going w/ nonparametric!



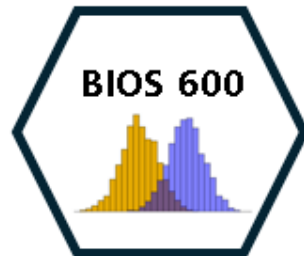
# When to Use Kruskal–Wallis vs. ANOVA

| Feature              | One-Way ANOVA                                                                           | Kruskal–Wallis Test                                                                             |
|----------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Purpose</b>       | Tests whether the <b>means</b> of 3+ groups differ                                      | Tests whether the <b>distributions</b> (typically medians) differ                               |
| <b>Data type</b>     | Continuous, roughly normal                                                              | Ordinal or continuous, not normal                                                               |
| <b>Assumptions</b>   | 1. Normality within groups 2. Equal variances (homoscedasticity) 3. Independent samples | 1. Independent samples 2. Same-shaped distributions (for median interpretation)                 |
| <b>Basis of test</b> | Uses <b>group means</b> and <b>within-group variance</b>                                | Uses <b>ranks</b> of data instead of raw values                                                 |
| <b>Example use</b>   | Compare mean blood pressure across 3 treatment groups                                   | Compare pain severity scores (on a 0–10 scale) across 3 treatment groups when scores are skewed |



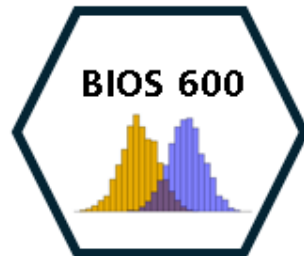
# Quick Intuition

- ▶ Use **ANOVA** if your data meet **parametric assumptions** (normal, equal variances).
- ▶ Use **Kruskal–Wallis** if your data are **skewed**, **ordinal**, or have **outliers** that violate ANOVA's assumptions.



# Why not always go nonparametric?

- ▶ Non-parametric methods are desirable because they do not require as many restrictive assumptions as parametric ones.
- ▶ However, this flexibility comes at a price – if the assumptions underlying a parametric test are satisfied, the nonparametric test is **less powerful** than the comparable parametric technique. We'll talk about statistical power next class.



# Next Class

- ▶ Power and sample size calculation (incredibly useful!)

